

Project Proposal

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Abstract

High-resolution spectroscopic surveys provide precise stellar parameters and chemical abundances, but their samples contain only a small fraction of the stars with astrometry and low-resolution spectrophotometry from *Gaia*. Building directly on the Gaia XP chemical-tagging framework developed by Kane et al., we retrain and adapt a multi-output neural network to transfer labels from APOGEE DR17 to *Gaia* DR3 BP/RP XP spectra for a targeted study of the Sagittarius stellar stream. We use the Sagittarius stream catalogue of ?] to define the target population, crossmatch its original *Gaia* DR2 sources to *Gaia* DR3, and obtain XP spectra and updated source measurements. The model predicts T_{eff} , $\log g$, $[\text{Fe}/\text{H}]$, $[\text{N}/\text{O}]$, and $[\text{Al}/\text{Fe}]$, together with a source-dependent variance for each label. We use a heteroscedastic Gaussian likelihood to model observational scatter and Monte Carlo dropout to estimate model uncertainty. This study will expand the number of Sagittarius stream stars with estimated stellar labels while also testing machine-learning methods for astronomy and astrophysics.

1 Introduction

The chemical and dynamical properties of stars preserve information about the formation of the Milky Way and its accreted satellites. In particular, metallicities and elemental-abundance ratios can distinguish stellar populations, constrain enrichment histories, and identify debris associated with disrupted dwarf galaxies. The Sagittarius dwarf spheroidal galaxy (hereafter Sgr) is a nearby satellite undergoing tidal disruption by the Milky Way and therefore provides a valuable laboratory for studying chemical evolution during an ongoing accretion event.

Detailed abundance measurements are commonly obtained from medium- and high-resolution spectroscopic surveys. The Apache Point Observatory Galactic Evolution Experiment (APOGEE) provides precise near-infrared stellar labels for hundreds of thousands of stars, including a spectroscopic subset of the Sgr system [AAA⁺22]. However, the number of probable Sgr stream members identified using *Gaia* astrometry is much larger than the number with existing spectroscopic metallicities and abundances. ?] assembled a catalogue of 55,192 high-probability Sagittarius stream candidates using *Gaia* DR2 astrometry, 2MASS photometry, distance estimates based on nearby RR Lyrae stars, and available line-of-sight velocities from several spectroscopic surveys. Only a subset of these candidates currently has detailed spectroscopic labels.

Gaia Data Release 3 (DR3) provides internally calibrated low-resolution BP/RP spectrophotometry, commonly referred to as XP spectra, for roughly 220 million sources [GV⁺23, DWM⁺23]. These data are represented by basis-function coefficients and contain broad spectral information across the optical and near-infrared wavelength range. Several recent studies have demonstrated that data-driven models can infer stellar parameters, metallicities, and selected abundance information from XP data [e.g., L⁺24, KAC⁺24]. The information available at XP resolution is nevertheless entangled with temperature, surface gravity, extinction, survey selection, and correlations among chemical labels. Consequently, both careful validation and a quantitative assessment of predictive uncertainty are required before applying a trained model to a distinct population such as the Sgr stream.

Of particular relevance, (author?) [KBC⁺25] developed a heteroscedastic multilayer perceptron that used *Gaia* BP/RP coefficients to infer T_{eff} , $\log g$, $[\text{Fe}/\text{H}]$, $[\text{N}/\text{O}]$, and $[\text{Al}/\text{Fe}]$ on the APOGEE abundance scale. Their network used 108 normalized BP/RP coefficients, two hidden layers of 128 neurons, dropout with $p = 0.2$, and 100 stochastic forward passes. They used the resulting high- $[\text{N}/\text{O}]$ and high- $[\text{Al}/\text{Fe}]$ predictions to identify 878 new candidate globular-cluster-origin stars in the Galactic field, reporting an overall classification accuracy of approximately 99% and a false-positive rate of

approximately 7%. This work demonstrated both the chemical-tagging potential of XP spectra and the importance of abundance-specific validation at low spectral resolution.

For every target label, the network will predict both a mean and an input-dependent variance. We additionally use dropout at inference time to estimate the sensitivity of the prediction to the learned network parameters [GG16, KG17, KBC+25]. We will apply the trained model to the Sagittarius stream candidates identified by [?], after updating the catalogue through crossmatching with *Gaia* DR3 and supplementing it with APOGEE DR17 spectroscopic labels where available.

The primary goals of this study are to:

1. determine the precision with which T_{eff} , $\log g$, $[\text{Fe}/\text{H}]$, $[\text{N}/\text{O}]$, and $[\text{Al}/\text{Fe}]$ can be transferred from APOGEE to stars with *Gaia* XP spectra;
2. test whether the predicted uncertainties are statistically calibrated;
3. identify the magnitude, color, stellar-parameter, and XP-coefficient domains in which the model provides reliable interpolation rather than extrapolation; and
4. use the expanded label sample to characterize variations in metallicity and chemical abundance along the Sagittarius stellar stream.

The paper is organized as follows. Section 2 describes the APOGEE–*Gaia* training sample, the [?] Sagittarius stream catalogue, and the catalogue crossmatching procedure. Section 2.4 presents the network architecture, loss function, training procedure, and uncertainty model. Section 3 validates the predictions on held-out stars. Section 4 presents the application to the Sagittarius stellar stream. We discuss the astrophysical interpretation and limitations in Section 5, and summarize our conclusions in Section 6.

2 Data

2.1 Gaia DR3 XP Spectra

Gaia DR3 publishes BP and RP mean spectra as coefficients multiplying internally calibrated basis functions [DWM+23]. For each source, we concatenate the retained BP and RP coefficients into a feature vector

$$\mathbf{x}_i = (c_{i,1}^{\text{BP}}, \dots, c_{i,N_{\text{BP}}}^{\text{BP}}, c_{i,1}^{\text{RP}}, \dots, c_{i,N_{\text{RP}}}^{\text{RP}}), \quad (1)$$

where i indexes the star. The input construction is adapted from (author?) [KBC+25]. In their preprocessing, each of the 55 BP and 55 RP coefficients is divided by the first coefficient of the corresponding arm; the two resulting unit-valued first coefficients are then removed, leaving 108 predictors. Our final pipeline therefore uses $N_{\text{XP}} = 108$ input features.

We retrieve XP coefficients from the appropriate *Gaia* DR3 XP continuous mean-spectrum table and associate them with source information from `gaia_source`. Sources with missing or non-finite coefficients are removed. We additionally investigate the dependence of model performance on G magnitude, $G_{\text{BP}} - G_{\text{RP}}$ color, extinction, and XP goodness-of-fit statistics.

2.2 APOGEE DR17 Labels

The reference labels are drawn from APOGEE DR17 [AAA+22]. Following the label-transfer problem defined by (author?) [KBC+25], we use the following five targets:

$$\mathbf{y}_i = (T_{\text{eff}}, \log g, [\text{Fe}/\text{H}], [\text{N}/\text{O}], [\text{Al}/\text{Fe}])_i. \quad (2)$$

The exact catalogue columns are `TEFF`, `LOGG`, `FE_H`, `NO`, and `AL_FE`. The label denoted $[\text{N}/\text{O}]$ throughout this paper must be verified against the data-reading function and APOGEE DR17 documentation before analysis and publication.

The general model-training sample will be constructed from stars that have both *Gaia* DR3 XP spectra and valid APOGEE DR17 labels. This broader APOGEE–*Gaia* overlap will provide substantially more training examples than the Sagittarius sample alone. The Sagittarius stars with existing APOGEE labels will be reserved for population-specific evaluation and comparison unless a later experiment explicitly tests their inclusion in the training data.

2.3 Sagittarius Stream Catalogue

The target Sagittarius stream sample will be drawn from the catalogue of ?]. The published catalogue contains 55,192 high-probability candidate members of the Sagittarius stellar stream. It focuses primarily on red giant branch stars, which are bright enough to provide relatively precise proper-motion measurements. The original catalogue combines positions and proper motions from *Gaia* DR2, near-infrared photometry from 2MASS, distance estimates based on nearby RR Lyrae stars, and line-of-sight velocities and metallicities from several spectroscopic surveys where available. Approximately 4,500 entries have line-of-sight velocity measurements.

Because the catalogue was constructed using *Gaia* DR2, we will crossmatch its Sagittarius candidates to *Gaia* DR3. We will retain the Sagittarius membership selection supplied by ?], but replace or supplement the original DR2 measurements with their DR3 counterparts where a reliable association is available. The DR3 match will provide updated astrometry and photometry as well as access to the BP/RP XP coefficients required by the neural network.

We will not assume that a DR2 source identifier is interchangeable with a DR3 source identifier. Wherever possible, we will use an official *Gaia* cross-release neighborhood or best-neighbor table to identify the corresponding DR3 source. Positional matching, with coordinates propagated to a common epoch when necessary, will be used as a secondary method. Matches will be screened using angular separation, consistency of proper motion and photometry, and the number of plausible counterparts. Ambiguous matches, sources exceeding the adopted match tolerance, and sources without usable DR3 XP coefficients will be removed or flagged for further inspection. The final match radius and quality thresholds will be reported in the completed analysis.

The DR3-updated Sagittarius sample will then be crossmatched with APOGEE DR17. Stars found in both datasets will provide independently measured APOGEE labels that can be compared with the network predictions. Sagittarius candidates without APOGEE labels, but with valid DR3 XP coefficients and acceptable quality measurements, will form the application sample for which the trained model will estimate the missing stellar parameters and chemical abundances.

The complete data flow is therefore: (1) train the model using the general APOGEE DR17–*Gaia* DR3 overlap; (2) update the ?] Sagittarius catalogue from DR2 to DR3; (3) identify Sagittarius candidates with existing APOGEE DR17 measurements for population-specific validation; and (4) apply the validated model to the remaining Sagittarius candidates with usable DR3 XP spectra.

2.4 Proposed Methodology

We propose to develop and train a multilayer perceptron (MLP) neural network to predict missing stellar labels for stars in the *Gaia* catalogue. The network will be trained as a supervised regression model using stars for which both *Gaia* DR3 XP observations and independently measured APOGEE DR17 stellar labels are available. After validation, the trained model will be applied to Sagittarius stream candidates lacking these labels in order to infer their stellar parameters and chemical abundances.

The input features will consist of the 108 normalized *Gaia* XP coefficients, while the target variables will be the corresponding known APOGEE labels in the training sample. Prior to training, the data will be cleaned, normalized, and divided into training, validation, and test sets. Preprocessing transformations will be fitted using only the training set to prevent information leakage. The validation set will be used to monitor model performance and select hyperparameters, while the test set will provide an independent assessment of predictive accuracy.

The MLP will consist of multiple fully connected hidden layers with nonlinear activation functions. For each stellar label, the output layer will predict both a mean and a variance. Model parameters will be optimized through backpropagation by minimizing a heteroscedastic Gaussian negative log-likelihood. We will investigate the effects of network architecture, learning rate, batch size, dropout probability, and other hyperparameters on model performance.

The model will be implemented primarily using PyTorch, with scikit-learn used for data preprocessing, dataset splitting, and performance evaluation. We plan to use Hugging Face Accelerate to manage PyTorch training on available GPU hardware, reducing training time and supporting efficient hyperparameter experiments. The final computing environment, software versions, selected hyperparameters, and random seeds will be documented to support reproducibility.

3 Validation

[TODO: Add held-out test results, residual analysis, and uncertainty calibration.]

4 Sagittarius Stream Results

[TODO: Add the application to the DR3-updated Vasiliev et al. catalogue.]

5 Discussion

[TODO: Discuss astrophysical interpretation, selection effects, and limitations.]

6 Conclusions

[TODO: Summarize the final results.]

References

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